

**SAT Math Mastery · Week TP1D****SAT Trick Pack 1 — Part D****Time** 120 minutes: 90 min solving + 30 min self-check and error log**Topics** exponential rates · angle bisectors · unit conversion · cylinders · confidence intervals · circles and tangency · parallel lines · spheres · quadratic intersections · altitude to hypotenuse · discriminants · proportional equations · pyramids · parametrised circles**Name / Date** \_\_\_\_\_**1 · SKILLS & IDEAS**

- S1.** Read the per-unit growth factor of an exponential model and convert it to a percent change (and back from a multi-unit change).
- S2.** Enumerate angle sums systematically from a small set of repeating angle types.
- S3.** Chain unit conversions so units cancel; square the factor when the unit is squared.
- S4.** Eliminate a dimension with a volume condition, solve the resulting cubic by testing integer roots, then apply the extra constraint.
- S5.** Interpret a confidence interval and scale the margin of error with  $1/\sqrt{n}$ .
- S6.** Use tangency (distance from center to line equals radius) to find a constant in a circle equation.
- S7.** Force parallel lines with equal slopes, then confirm the intercepts differ.
- S8.** Scale similar solids: radius ratio  $\rightarrow$  volume ratio by cubing.
- S9.** Write a quadratic as a line plus a multiple of the root factors when two graphs intersect.
- S10.** Use the altitude to the hypotenuse:  $h^2 = pq$ , then choose the right root and a trig ratio.
- S11.** Use the discriminant for “no intersection” and bracket a parameter between perfect squares.
- S12.** Recognise proportional equations (infinitely many solutions) and scale constants with the coefficients.
- S13.** Chain base  $\rightarrow$  side  $\rightarrow$  height  $\rightarrow$  slant height  $\rightarrow$  face area in a pyramid.
- S14.** Test whether a parametrised point stays on a parametrised circle by simplifying its offsets from the center.

**2 · REQUIRED FACTS**

- If  $f(x) = ab^{x/k}$ , then  $f(x+1)/f(x) = b^{1/k}$  and the percent change per unit is  $b^{1/k} - 1$  (retained percent =  $b^{1/k}$ ).
- A growth factor over  $n$  units is  $r^n$ ; the factor per unit is its  $n$ th root.
- A confidence interval is a range of plausible values, not certain ones; the margin of error is proportional to  $\frac{1}{\sqrt{n}}$ .
- A line is tangent to a circle exactly when the distance from the center to the line equals the radius.
- Two lines with equal slopes are parallel (no solution) if the intercepts differ and identical (infinitely many solutions) if they match.
- Spheres are similar: radii in ratio  $k$  give surface areas in ratio  $k^2$  and volumes in ratio  $k^3$ .
- Two graphs have equal outputs at their intersections, so the difference of the two functions has those  $x$ -values as roots.
- In a right triangle with altitude  $h$  to the hypotenuse splitting it into  $p$  and  $q$ :  $h^2 = pq$ .
- A system with no real solution leads to a quadratic with negative discriminant.
- Time appears squared in an acceleration; one length unit and two time units must be converted.

**3 · REQUIRED FORMULAS**

$$\text{percent change} = b^{1/k} - 1, \quad \frac{ME_2}{ME_1} = \sqrt{\frac{n_1}{n_2}}, \quad d = \frac{|Ah + Bk + C|}{\sqrt{A^2 + B^2}}, \quad V_{\text{cyl}} = \pi r^2 h, \quad S_{\text{cyl}} = 2\pi r^2 + 2\pi r h$$

$$\Delta = b^2 - 4ac, \quad V_{\text{pyr}} = \frac{1}{3}Bh, \quad \ell^2 = h^2 + \left(\frac{s}{2}\right)^2, \quad A_{\text{lat}} = 4 \cdot \frac{1}{2}s\ell, \quad S_{\text{sph}} = 4\pi r^2, \quad V_{\text{sph}} = \frac{4}{3}\pi r^3$$

#### 4 · MINI THEORY SHEET

**Per-unit factors.** Compare outputs one input unit apart, or rewrite  $b^{x/k} = (b^{1/k})^x$ . Percent change = factor  $-1$ ; do not report the retained percent.

**Chained conversions.** Write the rate as a fraction, multiply by conversion fractions so every unwanted unit cancels, then scale to the requested quantity. Squared units square the factor.

**Lines and curves.** If two graphs meet at  $x = r, s$ , then  $p(x) = L(x) + a(x - r)(x - s)$ ; one more point fixes  $a$ .

**Constraints on roots.** A cubic from a volume condition usually has small integer roots; test divisors of the constant term, then use the stated inequality to pick the valid one.

**Parametrised objects.** Substitute the parametrised point and simplify; a parameter that cancels leaves a statement that is true or false for all values.

#### Speed shortcuts.

| Situation                                | Shortcut                                                                                |
|------------------------------------------|-----------------------------------------------------------------------------------------|
| $f(x) = ab^{x/k}$                        | factor per unit is $b^{1/k}$ ; percent = factor $-1$                                    |
| growth over $n$ units                    | per-unit factor = ( $n$ th root of total factor)                                        |
| rotating bisectors                       | list gaps ( $\frac{x}{2}$ and $90 - \frac{x}{2}$ ), add consecutive ones                |
| rate in mi/gal, want L for km            | invert the rate, chain the factors                                                      |
| cylinder, volume and surface given       | substitute $h = \frac{V}{\pi r^2}$ , test small integer $r$                             |
| margin of error                          | $\propto \frac{1}{\sqrt{n}}$ : four times the sample halves it                          |
| circle with tangent line                 | distance from center to line = $r$                                                      |
| $kx + 3y$ vs $4x + (k - 1)y$             | equal slopes, then check the constants                                                  |
| sphere, circumference and area           | radius ratio, then cube                                                                 |
| parabola meets line at $r, s$            | $p = L + a(x - r)(x - s)$ , solve $a$ from one point                                    |
| altitude to hypotenuse                   | $h^2 = pq$ ; choose root by the inequality                                              |
| parabola and line, no intersection       | $\Delta < 0$ ; bracket between perfect squares                                          |
| proportional equations                   | scale factor from coefficients, apply to the constant                                   |
| pyramid                                  | diagonal $\rightarrow$ side $\rightarrow$ height $\rightarrow$ slant $\rightarrow$ face |
| point on parametrised circle             | simplify offsets; parameters must cancel                                                |
| $\text{m/s}^2 \rightarrow \text{km/h}^2$ | multiply by $3600^2/1000$                                                               |

#### 5 · GUIDED EXAMPLES

##### Example 1 — Squared time unit

A cart accelerates at  $0.2 \text{ m/s}^2$ . Express this in  $\text{km/h}^2$ .

$$0.2 \frac{\text{m}}{\text{s}^2} \cdot \frac{1 \text{ km}}{1000 \text{ m}} \cdot \left(\frac{3600 \text{ s}}{1 \text{ h}}\right)^2 = 0.2 \cdot \frac{12,960,000}{1000} = 2592 \text{ km/h}^2.$$

##### Example 2 — Line plus factored term

A quadratic  $p$  and the line  $L(x) = x + 2$  intersect at  $x = 1$  and  $x = 5$ , and  $p(0) = -3$ . Find  $p(3)$ .

$p(x) = x + 2 + a(x - 1)(x - 5)$ . At 0:  $2 + 5a = -3 \Rightarrow a = -1$ . Then  $p(3) = 5 - (2)(-2) = 9$ .

##### Example 3 — No intersection means $\Delta < 0$

For which integers  $m$  do  $y = x^2 + 2x + 5$  and  $y = mx + 1$  have no common point?

$x^2 + (2 - m)x + 4 = 0$ ,  $\Delta = (2 - m)^2 - 16 < 0 \Rightarrow |m - 2| < 4 \Rightarrow -2 < m < 6$ . Integers:  $-1, 0, \dots, 5$  (and  $m = 6$  gives a tangent line, not allowed).

## 6 · PRACTICE SET — 16 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. SAT CHALLENGE · CALCULATOR

The value  $V$ , in dollars, of a delivery van  $x$  years after it is purchased is modeled by  $V(x) = 18,000(0.49)^{x/2}$ . Which of the following best describes how the van's value changes each year?

(A) It decreases by 24.5% each year.

(B) It decreases by 30% each year. ✓

(C) It decreases by 70% each year.

(D) It decreases by 51% each year.

1. Rewrite the exponent:  $(0.49)^{x/2} = \left((0.49)^{1/2}\right)^x = (0.7)^x$ .

2. So  $V(x) = 18,000(0.7)^x$  and each year the value is multiplied by 0.7.

3. The van keeps 70% of its value, so it loses  $1 - 0.7 = 0.30$ , i.e. 30%, each year.

4. Check:  $\frac{V(3)}{V(2)} = \frac{(0.49)^{1.5}}{(0.49)^1} = 0.49^{0.5} = 0.7$ . The option 51% is  $1 - 0.49$  (the two-year loss).

Answer: (B)

2. SAT CHALLENGE · NO CALCULATOR NEEDED

Two lines intersect at point  $O$ , and the smallest of the four angles they form measures  $50^\circ$ . A ray from  $O$  bisects each of the four angles. Together with the two lines, this gives eight rays from  $O$ , which divide the full turn into eight small angles. A *span* is an angle formed by 1, 2 or 3 consecutive small angles. Which of the following is NOT the measure of any span?

(A)  $130^\circ$

(B)  $155^\circ$

(C)  $140^\circ$  ✓

(D)  $115^\circ$

1. The lines cut the plane into angles of  $50^\circ$  and  $130^\circ$ . Bisecting gives pieces of  $25^\circ$  and  $65^\circ$ .

2. Going around  $O$  the small angles are 25, 25, 65, 65, 25, 25, 65, 65 (degrees), repeating.

3. One small angle: 25 or 65. Two consecutive:  $25+25 = 50$ ,  $25+65 = 90$ ,  $65+65 = 130$ . Three consecutive:  $25+25+65 = 115$ ,  $25+65+65 = 155$  (and the same sums in the other order).

4. The possible spans are 25, 65, 50, 90, 130, 115, 155. The values 115, 130 and 155 occur, but 140 does not.

Answer: (C)

3.

SAT CHALLENGE · CALCULATOR

A car's fuel economy is 40 miles per gallon. Use 1 mile = 1.6 kilometers and 1 gallon = 3.8 liters. How many liters of fuel does the car need to drive 304 kilometers?

1. Convert the rate to kilometers per gallon:  $40 \cdot 1.6 = 64$  km per gallon.

2. Gallons needed:  $\frac{304}{64} = 4.75$  gallons.

3. Liters:  $4.75 \cdot 3.8 = 18.05$ .

4. (Equivalently invert first:  $\frac{3.8 \text{ L}}{64 \text{ km}} \cdot 304 \text{ km} = 18.05 \text{ L.}$ )

**Answer:** 18.05

4.

SAT CHALLENGE · CALCULATOR

A closed right circular cylinder has a volume of  $48\pi$  cubic centimeters and a total surface area of  $56\pi$  square centimeters. The height of the cylinder is greater than its diameter. What is the height, in centimeters, of the cylinder?

(A) 12 ✓

(B) 2

(C) 3

(D) 4

1. From  $\pi r^2 h = 48\pi$ :  $h = \frac{48}{r^2}$ .

2. Surface area:  $2\pi r^2 + 2\pi r h = 56\pi \Rightarrow r^2 + rh = 28 \Rightarrow r^2 + \frac{48}{r} = 28$ .

3. Multiply by  $r$ :  $r^3 - 28r + 48 = 0$ . Test divisors of 48:  $r = 2$  gives  $8 - 56 + 48 = 0$ ,  $r = 4$  gives  $64 - 112 + 48 = 0$ . Factor:  $(r - 2)(r - 4)(r + 6) = 0$ , so the positive radii are 2 and 4.

4.  $r = 2$ :  $h = 12 > 4 = 2r$ .  $r = 4$ :  $h = 3 < 8 = 2r$ , which violates the condition.

5. The height is 12 centimeters. (The other choices are the radii 2 and 4 and the height 3 of the rejected solution.)

**Answer:** (A)

5.

SAT CHALLENGE · CALCULATOR

A researcher surveys a random sample of 625 workers in a city and calculates a 95% confidence interval of 24.2 to 27.8 minutes for the mean one-way commute time of all workers in the city. Assume a second random sample of 2,500 workers has similar variability and the same method is used. Which of the following statements must be true?

(A) About 95% of sampled commutes lie in the interval. (B) The new margin is about 0.45 min.

(C) The population mean cannot be 28.1 min.

**(D) The new margin is about 0.9 min. ✓**

1. The interval's center is 26 and its margin of error is  $\frac{27.8 - 24.2}{2} = 1.8$  minutes.
2. Margin of error is proportional to  $\frac{1}{\sqrt{n}}$ . The sample is 4 times as large, so the margin is divided by  $\sqrt{4} = 2$ :  $1.8 \div 2 = 0.9$  minutes. This matches the statement about 0.9 minutes.
3. The statement that the mean cannot be 28.1 is false: a confidence interval gives plausible values; it does not make 28.1 impossible. The statement about 95% of sampled commutes is false: the interval is about the population mean, not the middle 95% of individual commutes. The 0.45 statement divides by 4 instead of  $\sqrt{4}$ .

**Answer: (D)**

6.

SAT CHALLENGE · CALCULATOR

The area covered by a coral colony increases by 33.1% every 3 years. If the colony grows by the same percent each year, by what percent does its area increase each year?

1. Over 3 years the area is multiplied by 1.331. If the yearly factor is  $b$ , then  $b^3 = 1.331$ .
2. Take the cube root:  $b = \sqrt[3]{1.331} = 1.1$ , because  $1.1^3 = 1.331$ .
3. A factor of 1.1 is an increase of  $1.1 - 1 = 0.10$ , that is 10% per year.

**Answer: 10**

7.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$x^2 + y^2 - 6x + 8y = k$$

In the  $xy$ -plane, the graph of the equation above, where  $k$  is a constant, is a circle that is tangent to the line  $4x - 3y = -11$ . What is the value of  $k$ ?

1. Complete the square:  $(x - 3)^2 + (y + 4)^2 = k + 25$ . The center is  $(3, -4)$  and  $r^2 = k + 25$ .

2. Distance from the center to  $4x - 3y + 11 = 0$ :  $\frac{|4(3) - 3(-4) + 11|}{\sqrt{16 + 9}} = \frac{35}{5} = 7$ .

3. Tangency means  $r = 7$ , so  $r^2 = 49 = k + 25$ .

4.  $k = 24$ .

**Answer:** 24

8.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$kx + 3y = 12$$

$$4x + (k - 1)y = 12$$

In the system of equations above,  $k$  is a constant. For which value of  $k$  does the system have no solution?

(A) 3

(B) -4

(C) -3 ✓

(D) 4

1. No solution requires parallel, distinct lines: equal slopes. Slopes are  $-\frac{k}{3}$  and  $-\frac{4}{k-1}$ .

2.  $\frac{k}{3} = \frac{4}{k-1} \Rightarrow k(k-1) = 12 \Rightarrow k^2 - k - 12 = 0 \Rightarrow (k-4)(k+3) = 0$ .

3.  $k = 4$ : the equations are  $4x + 3y = 12$  and  $4x + 3y = 12$ , the same line, so infinitely many solutions.

4.  $k = -3$ :  $-3x + 3y = 12$  gives  $y = x + 4$ , and  $4x - 4y = 12$  gives  $y = x - 3$ . Parallel and distinct, so no solution.

5. The answer is  $k = -3$ .

**Answer:** (C)

9.

SAT CHALLENGE · NO CALCULATOR NEEDED

Sphere  $A$  has a great circle with circumference  $8\pi$  centimeters. The surface area of sphere  $B$  is  $80\pi$  square centimeters greater than the surface area of sphere  $A$ . What is the ratio of the volume of sphere  $B$  to the volume of sphere  $A$ ?

(A)  $\frac{9}{4}$

(B)  $\frac{27}{8}$  ✓

(C)  $\frac{5}{4}$

(D)  $\frac{3}{2}$

1. A great circle of sphere  $A$  has circumference  $2\pi r_A = 8\pi$ , so  $r_A = 4$  and its surface area is  $4\pi(16) = 64\pi$ .

2. Sphere  $B$ :  $4\pi r_B^2 = 64\pi + 80\pi = 144\pi \Rightarrow r_B^2 = 36 \Rightarrow r_B = 6$ .

3. Radius ratio  $\frac{6}{4} = \frac{3}{2}$ . Volumes scale with the cube:  $\left(\frac{3}{2}\right)^3 = \frac{27}{8}$ .

4. (Other choices:  $\frac{3}{2}$  is the radius ratio,  $\frac{9}{4}$  is the area ratio, and  $\frac{5}{4} = \frac{80}{64}$  compares the area difference with sphere  $A$ 's area.)

**Answer: (B)**

10.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the  $xy$ -plane, the graph of the quadratic function  $p$  and the graph of the line  $y = 5 - x$  intersect at exactly two points, one with  $x$ -coordinate 1 and one with  $x$ -coordinate 4. If  $p(0) = -3$ , what is the value of  $p(6)$ ?

1. At the intersections  $p(x) = 5 - x$ , so  $p(x) - (5 - x)$  has zeros 1 and 4. Write  $p(x) = 5 - x + a(x - 1)(x - 4)$ .

2. Use  $p(0) = -3$ :  $5 + a(-1)(-4) = -3 \Rightarrow 4a = -8 \Rightarrow a = -2$ .

3.  $p(6) = 5 - 6 - 2(5)(2) = -1 - 20 = -21$ .

4. (Check by expanding:  $p(x) = -2x^2 + 9x - 3$ , and  $p(6) = -72 + 54 - 3 = -21$ .)

**Answer: -21**

11.

SAT CHALLENGE · CALCULATOR

In right triangle  $ABC$ , the right angle is at  $C$  and  $AB = 10$ . Point  $D$  lies on  $\overline{AB}$  with  $\overline{CD} \perp \overline{AB}$  and  $CD = 4$ . If  $AD > DB$ , what is the value of  $\cos A$ ?

(A)  $\frac{\sqrt{5}}{5}$

(B)  $\frac{4}{5}$

(C)  $\frac{1}{2}$

(D)  $\frac{2\sqrt{5}}{5}$  ✓

1. Let  $AD = p$  and  $DB = q$ . Then  $p + q = 10$ , and the altitude to the hypotenuse gives  $CD^2 = pq = 16$ .

2.  $p(10 - p) = 16 \Rightarrow p^2 - 10p + 16 = 0 \Rightarrow (p - 2)(p - 8) = 0$ . Since  $AD > DB$ ,  $AD = 8$  and  $DB = 2$ .

3. In right triangle  $ACD$ :  $AC = \sqrt{8^2 + 4^2} = \sqrt{80} = 4\sqrt{5}$ .

4.  $\cos A = \frac{AD}{AC} = \frac{8}{4\sqrt{5}} = \frac{2}{\sqrt{5}} = \frac{2\sqrt{5}}{5}$ .

5. (Other choices:  $\frac{\sqrt{5}}{5} = \sin A$ ,  $\frac{4}{5} = \frac{AD}{AB}$ ,  $\frac{1}{2} = \tan A$ .)

**Answer: (D)**

12.

SAT CHALLENGE · DESMOS

In the  $xy$ -plane, the graphs of  $y = x^2 + 6x + 4$  and  $y = mx - 4$  have no point in common, where  $m$  is a constant. What is the sum of all integer values of  $m$ ?

1. Set the expressions equal:  $x^2 + 6x + 4 = mx - 4 \Rightarrow x^2 + (6 - m)x + 8 = 0$ .

2. No common point means no real roots:  $\Delta = (6 - m)^2 - 32 < 0$ , so  $|m - 6| < \sqrt{32} \approx 5.66$ .

3. Since  $5^2 = 25 < 32 < 36 = 6^2$ ,  $|m - 6| \leq 5$ , i.e.  $1 \leq m \leq 11$ . (At  $m = 0$  or  $m = 12$  we get  $|m - 6| = 6$ , which is too large.)

4. Integers 1 through 11 sum to  $\frac{11 \cdot 12}{2} = 66$ .

**Answer: 66**

13.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$6x - 2(y - 5) = 2x + 3y + 1$$

$$ax + 20y = b$$

In the system above,  $a$  and  $b$  are constants. If the system has infinitely many solutions, what is the value of  $\frac{b}{a}$ ?

1. Simplify the first equation:  $6x - 2y + 10 = 2x + 3y + 1 \Rightarrow 4x - 5y = -9$ .
2. Infinitely many solutions means the second equation is a multiple of the first. The  $y$ -coefficients are  $-5$  and  $20$ , so the multiplier is  $\frac{20}{-5} = -4$ .
3. Then  $a = -4 \cdot 4 = -16$  and  $b = -4 \cdot (-9) = 36$ .
4.  $\frac{b}{a} = \frac{36}{-16} = -\frac{9}{4} = -2.25$ .

**Answer:**  $-9/4$

14.

SAT CHALLENGE · CALCULATOR

A right pyramid has a square base. The diagonal of the base is  $10\sqrt{2}$  centimeters and the volume of the pyramid is 400 cubic centimeters. What is the lateral surface area, in square centimeters, of the pyramid (the four triangular faces, not the base)?

(A) 260 ✓

(B) 240

(C) 130

(D) 360

1. The base diagonal is  $s\sqrt{2} = 10\sqrt{2}$ , so the side is  $s = 10$  and the base area is 100.
2. Volume  $\frac{1}{3}(100)h = 400 \Rightarrow h = 12$ .
3. The slant height (apex to the midpoint of a base edge) is the hypotenuse of a right triangle with legs  $h = 12$  and  $\frac{s}{2} = 5$ :  
 $\ell = \sqrt{144 + 25} = 13$ .
4. Each face has area  $\frac{1}{2}(10)(13) = 65$ , so the lateral area is  $4 \cdot 65 = 260$ .
5. (Using the height 12 instead of 13 gives 240; two faces give 130; adding the base gives 360.)

**Answer:** (A)

15.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$(x - a - 2)^2 + (y - 2a + 1)^2 = 25$$

In the  $xy$ -plane, the circle above is defined for each real number  $a$ . Which of the following points lies on the circle for every real value of  $a$ ?

(A)  $(a - 1, 2a + 2)$ **(B)**  $(a + 5, 2a + 3)$  ✓(C)  $(a + 5, 2a + 4)$ (D)  $(2a + 5, 2a + 3)$ 

1. The circle has center  $(a + 2, 2a - 1)$  and radius 5. For a point  $(x, y)$ , compute the offsets  $x - (a + 2)$  and  $y - (2a - 1)$  and see whether the  $a$ -terms cancel.
2.  $(a + 5, 2a + 3)$ : offsets 3 and 4, and  $3^2 + 4^2 = 25$ . ✓
3.  $(a + 5, 2a + 4)$ : offsets 3, 5, sum of squares 34.  $(2a + 5, 2a + 3)$ : first offset  $a + 3$  still depends on  $a$ .  $(a - 1, 2a + 2)$ : offsets  $-3, 3$ , sum of squares 18.
4. Only  $(a + 5, 2a + 3)$  works.

**Answer:** (B)

16.

SAT CHALLENGE · CALCULATOR

A commuter train accelerates at a constant rate of 0.5 meters per second squared. What is this acceleration, in kilometers per hour squared? (1 kilometer = 1000 meters and 1 hour = 3600 seconds.)

1. Write the acceleration with units:  $0.5 \frac{\text{m}}{\text{s}^2}$ . Meters become kilometers once: multiply by  $\frac{1 \text{ km}}{1000 \text{ m}}$ .
2. Seconds appear twice (squared), so the time factor must be applied twice: multiply by  $\left(\frac{3600 \text{ s}}{1 \text{ h}}\right)^2$ .
3.  $0.5 \cdot \frac{1}{1000} \cdot 3600^2 = 0.5 \cdot \frac{12,960,000}{1000} = 0.5 \cdot 12,960 = 6480$ .
4. The acceleration is 6480 km/h<sup>2</sup>.

**Answer:** 6480